higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 3

7 MARCH 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1
1.1.1 True
1.1.2 True
1.1.3 False
1.1.4 False
1.1.5 True
(5 × 1) (5)

1.2
1.2.1 C
1.2.2 B
1.2.3 D
1.2.4 A
(4 × 1) (4)

1.3
- Work overload
- Sexual harassment
- Discrimination
- Job insecurity
- Shift work
- Threat of workplace violence
- Lack of input or control (Any 3 × 1) (3)

1.4
- Inspect premises or workplace anytime without notice.
- Question any person in a workshop or premises.
- Ask to see any data or record book available.
- Examine any record book or documentation available.
- Ask questions relating to the record book.
- Inspect any equipment or machinery.
- Seize any record book or information for evidence.
- Ask any employer or employee to meet with him/her for questioning.
- Use an interpreter or SAP member for assistance. (Any 2 × 1) (2)

1.5 1.5.1
- Keep the working environment as safe as possible.
- Maintain all equipment and machinery.
- Provide well-maintained PPE.
- Train employees in safety and the safe operation of equipment.
- Keep workers informed.
- May not instruct employees to perform illegal activities.
- All work must be done under the supervision of trained people.

(Any 3 × 1) (3)

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1.5.2
- Must be responsible for their safety as well as that of others.
- Will not perform any unlawful activities even if instructed to do so.
- Must report any unsafe act or condition.
- Must cooperate with the employer and fellow workers.
- Must report any incident or accident as soon as possible.
- Must cooperate with any investigation that he/she has witnessed.

(Any 3 × 1) (3)

1.6
- Identify causes.
- Use correct procedures when collecting information or evidence.
- Record information as soon as possible, because circumstances can change.
- Collect physical evidence.
- Get eyewitness accounts.
- Discuss how similar incidents can be avoided in future. (Any 2 × 1) (2)

1.7
- Performing the same motion over and over.
- Physical effort such as pushing, pulling and lifting.
- Prolonged contact with hard, sharp surfaces.
- Working with vibrating tools.
- Straining the body such as bending, stooping, reaching or being in one position for long periods.
- Using dangerous substances without training.
- Operating faulty or unfamiliar machinery or equipment. (Any 3 × 1) (3)

[25]

QUESTION 2

2.1
- Magnetic effect
- Heating effect
- Chemical effect
- Electromagnetic effect

(Any 2 × 1) (2)

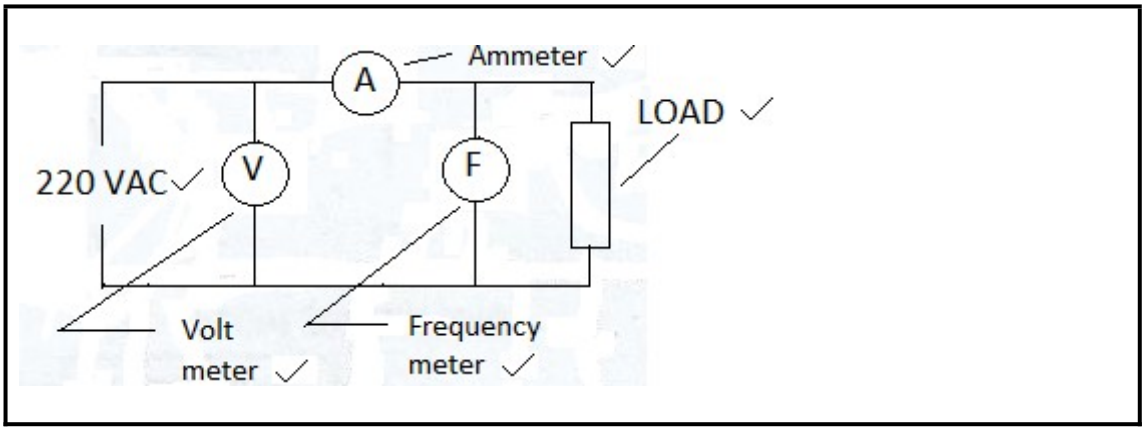
2.2
2.2.1 An earth-leakage/polarity tester
2.2.2 Clamp-on meter or ammeter
2.2.3 Voltmeter

(3 × 1) (3)

2.3
- Avoid rough handling.
- Do not drop instruments.
- Check for damaged or bent probes or cables.
- Allow instruments to warm up if necessary.
- Store in dry, clean, dust-free areas in suitable enclosures or casings provided, or on shelves.
- Select the correct range and scale.
- Check the instrument's operation before use.
- Do not apply more than the rated voltage or current.
- Disconnect the power and discharge all circuit capacitance.
- Remove the batteries if not used for long periods.
- Clean instruments after use.

(Any 2 × 1) (2)

2.4



(1 mark per correctly connected test instrument, supply and load) (5)

2.5 $\frac{Np}{Ns} = \frac{Vp}{Vs}$

$Vp = \frac{Np \times Vs}{Ns}$ ✓

$Vp = \frac{1\ 900 \times 20}{100}$

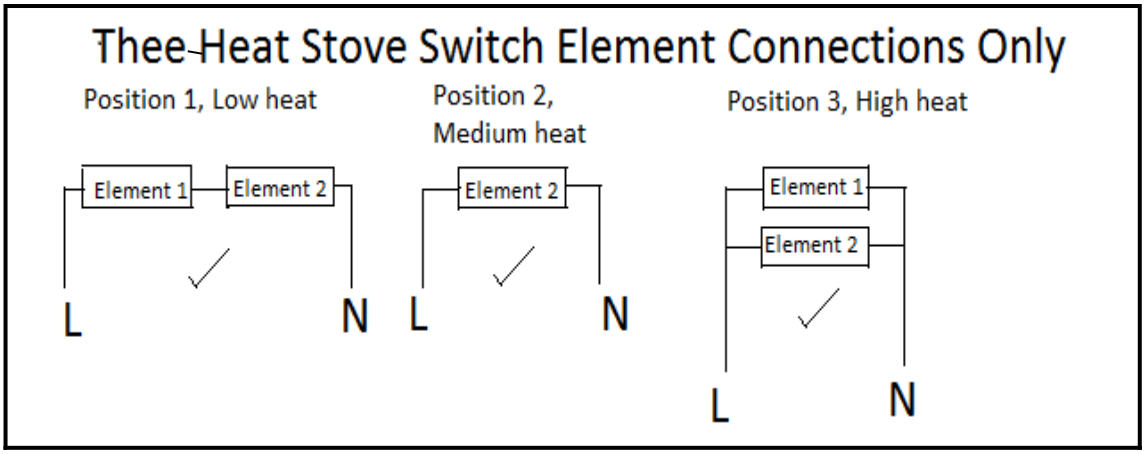
$Vp = \frac{38\ 000}{100}$ ✓

Vp = 380 V✓ (1 mark for the formula and 2 marks for the calculation) (3)
[15]

QUESTION 3

3.1
- The element heats up the oil causing it to become buoyant.
- The hot oil creates convection currents which move through the heater body over the heating element.
- The heat is then transferred through the metal casing to the surrounding area. (3)

3.2



(1 mark per complete circuit diagram) (3)

3.3
- A kettle has a bimetallic strip or contact which is mechanically connected to the on/off switch.
- When the power is switched on the element heats up the water and the pilot light switches on.
- When the water reaches boiling point the steam is channelled over the bimetallic strip which bends, forcing the switch off. (3)

3.4 3.4.1 This is an n/o switch which closes when the door or lid is closed completing the electrical circuit.

3.4.2 It regulates the oven temperature according to the selected temperature setting.

3.4.3 It operates by bending when it reaches a predetermined temperature, which then opens the contact.

(3 × 1) (3)

3.5
- These isolating devices must be easily accessible.
- They must be installed within arm's reach of the appliance.
- They must isolate the live and neutral for geysers and single-phase supply stoves. (3)

[15]

QUESTION 4

4.1
- Quality assurance
- Diagnostic testing
- Periodic safety checks
- Maintenance of equipment
- When performing fault finding on equipment (Any 2 × 1) (2)

4.2
- Basic insulation
- Double insulation
- Reinforced insulation
- Supplementary insulation (4)

4.3 4.3.1 F
4.3.2 D
4.3.3 A
4.3.4 G

(4 × 2) (8)

4.4
- Preassessment
- Observation
- Locate the fault
- Find the cause
- Rectify the fault (Any 1 × 1) (1)

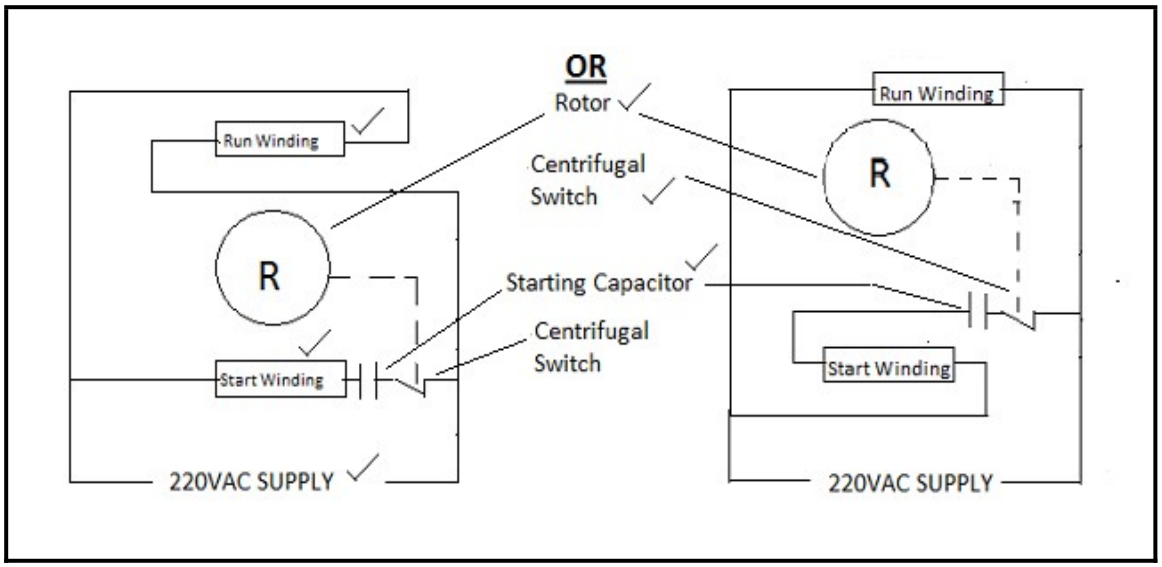
[15]

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QUESTION 5

5.1
- Ball bearings
- Tapered roller bearings
- Cylindrical roller bearings
- Needle roller bearings
- Sleeve bearings (Any 1 × 1) (1)

5.2



(1 mark per correctly connected component, symbol and label)
(Any other correct component and symbols are acceptable) (6)

5.3
- Series resistance or applied voltage control
- Tapped field method
- Centrifugal switch method (Any 2 × 1) (2)

5.4
- Air is drawn into the motor from a dust-free area via a sealed system.
- An internal fan cools the motor.
- Used in a dusty environment like carpenter workshops. (3)

5.5 5.5.1 Direct-on-line (1)

5.5.2
- The supply is switched directly across the motor.
- Used for small motors with high starting torque providing overload protection. (2 × 1) (2)

[15]

QUESTION 6

6.1
- Better for welding thin materials
- Better safety in damp, dangerous conditions
- Has a choice of polarities
- Light and portable (Any 2 × 1) (2)

6.2 6.2.1 The process of cleaning slag from the bead or welded joint done with a chipping hammer

6.2.2. The shape or outline of the face of the weld

6.2.3 The total angle between the groove face of one piece of metal and the groove face of the other

(3 × 1) (3)

6.3
- Forms a protective shield around the arc pool neutralising the surrounding air containing unwanted gases
- Impurities float to the top of the joint forming slag
- The slag seals the joint allowing it to cool down slowly
- Keeps the joint clean during the welding process
- Allows the molten metal to flow easily by directing the arc
- Influences the depth of the arc i.e. penetration (Any 2 × 1) (2)

6.4
- Place the point of the electrode at the start of the weld.
- Gently strike the metal lowering the electrode to form an arc.
- Lift the electrode 3 mm to 4 mm.
- Move the electrode in one direction along the joint towards yourself. (4)

6.5
- Chipping hammer
- Wire brush (2)

6.6
- The type of metal used or base metal
- The joint design
- The type of electrode used
- The welding position
- Pre-heating and post-heating requirements
- The welding machine settings
- Testing requirements
- Thickness of metal
- Groove design
- Joint alignment
- Welder skills
- Penetration depth
- Weld bead finish (Any 2 × 1) (2)

[15]

TOTAL: 100

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